MA3071 - DLI
Financial Mathematics – Section 5
Monte-Carlo methods for option pricing

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Background

- Option pricing is one of the key problems encountered in financial markets, and the Black–Scholes model is the most widely used model in practice.
- Despite its success in the pricing of many common types of options, the Black-Scholes model suffers computational intractability for many more exotic and complicated products in financial markets.
- Pricing such products is typically not practical using analytical methods, either because the level of mathematics required is far beyond that reasonably expected within a financial institution, or it is simply impossible using any of today's known techniques.

Background

- Hence computer models using Monte Carlo methods become increasingly important in these cases.
- The Monte Carlo method is a class of computational algorithms based on repeated random sampling to compute their results.
- It provides probabilistic algorithms for simulating systems where underlying randomness exists and is particularly powerful for high-dimensional problems that typically arise in finance.

Starter

- How to calculate the following expectations?
  - $\mathbb{E}[\sin(B_t)|\mathcal{F}_s]$.
  - $\mathbb{E}[(B_t - 1)^{B_t}|\mathcal{F}_s]$.
  - $\mathbb{E}[\cos(S_T)|\mathcal{F}_t]$ when S_t is a GBM.
- Do you think our conclusions still apply in the above cases?

Motivation

- In a binomial tree model, we can generate all possible sample paths for asset prices from time 0 to the maturity date.
- For example, consider a 2-period binomial tree. There are 4 possible paths from $S_0 \to S_2$.

$$
\begin{array}{ccccccc}
 & & & & S_2, & C_1, & \text{with } q_1 \\
 & & & \nearrow u & & & \\
 & & S_1 & & & & \\
 & \nearrow u & & \searrow d & S_2, & C_2, & \text{with } q_2 \\
S_0 & & & & & & \\
 & \searrow d & & \nearrow u & S_2, & C_3, & \text{with } q_3 \\
 & & S_1 & & & & \\
 & & & \searrow d & & & \\
 & & & & S_2, & C_4, & \text{with } q_4
\end{array}
$$

- Let the option payoff be $C = f(S_2)$. Then the expected payoff is

$$\mathbb{E}(C) = q_1 C_1 + q_2 C_2 + q_3 C_3 + q_4 C_4$$

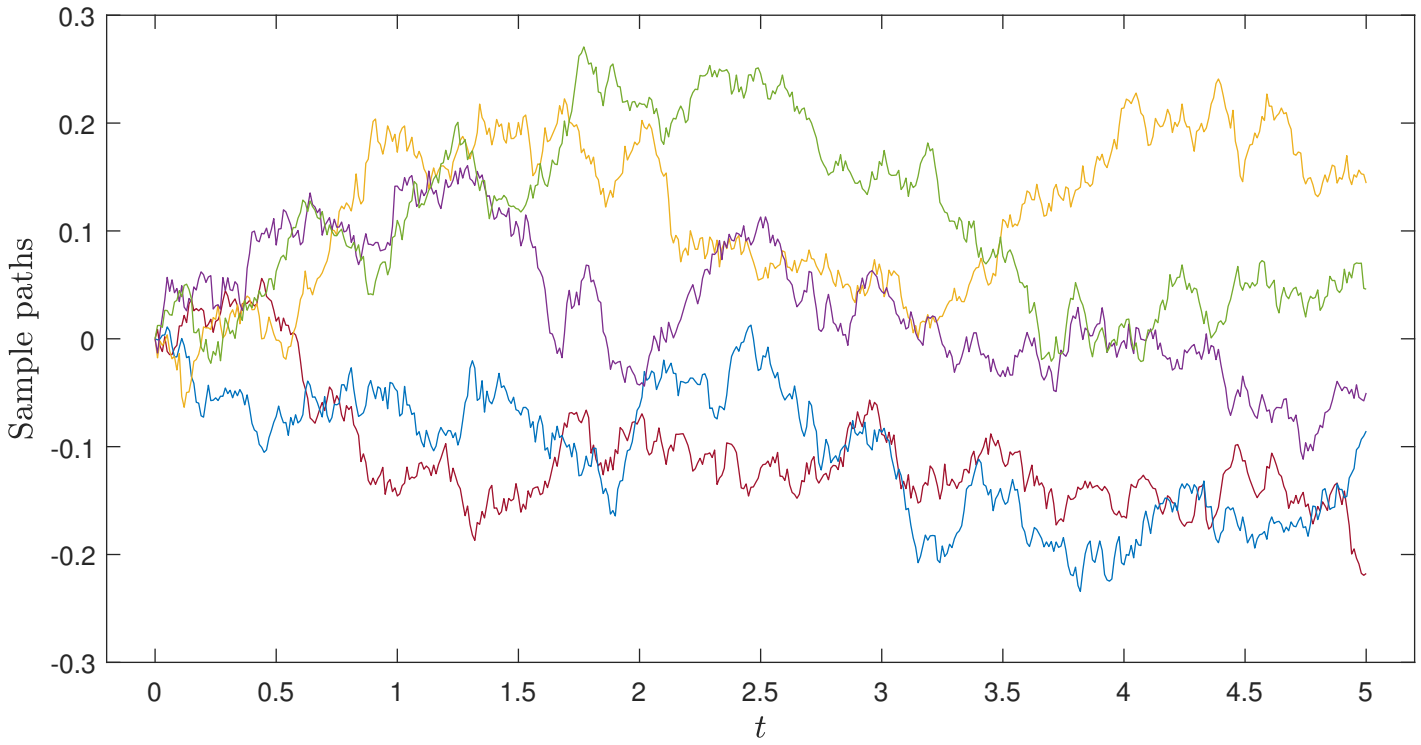
Motivation, cont.

- For an n-period binomial tree, the number of sample paths for $S_0 \to S_T$ is 2^n and therefore, we have 2^n possible values of the payoff.
- Let the option payoff be $C = f(S_T)$. Then the expected payoff is

$$\mathbb{E}(C) = \sum_{i=1}^{2^n} q_i C_i$$

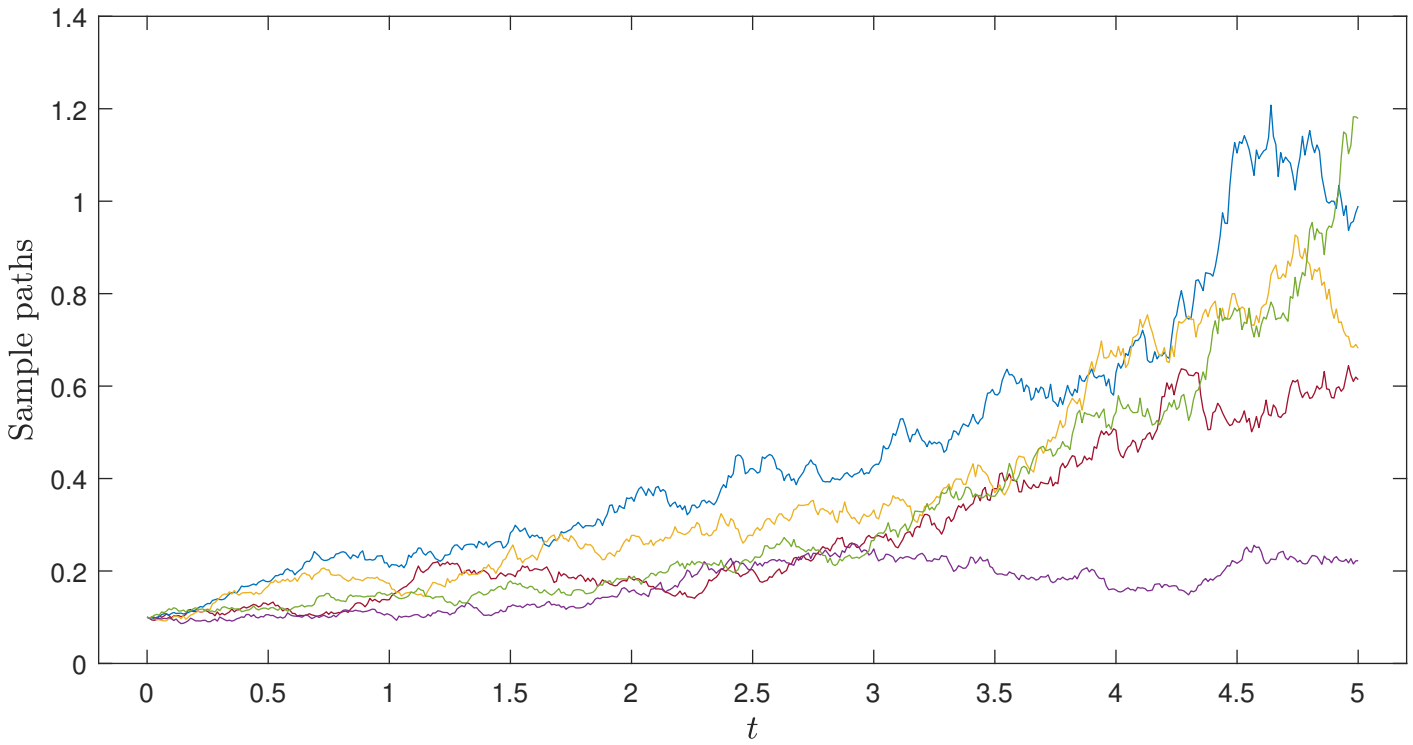
Sample paths for SBM

- We generate 5 sample paths for $B_0 \to B_5$,



Sample paths for GBM

- Let S_t be defined by $dS_t = \mu S_t dt + \sigma S_t dB_t$ with $\mu = 5$ and $\sigma = 3$. When $S_0 = 0.1$, we generate 5 sample paths for $S_0 \to S_5$,



Estimate of expectation

- Obviously, we cannot draw all possible sample paths for SBM or GBM, as B_t has an infinite number of possible values.
- However, as long as we generate enough different sample paths, we can still estimate expectations for SMB or GBM, such that
  - $\mathbb{E}[B_t] \approx \frac{1}{M} \sum_{i=1}^{M} B_t^{(i)}$
  - $\mathbb{E}[S_t] \approx \frac{1}{M} \sum_{i=1}^{M} S_t^{(i)}$

  where M is a large positive integer, $B_t^{(i)}$ and $S_t^{(i)}$ are observations of B_t and S_t, respectively.

Question

- How to collect observations for B_t and S_t?

Observations

- Recall that $B_t \sim N(0, t)$, than we have

$$\frac{B_t}{\sqrt{t}} \sim N(0, 1)$$

- Let z be an observation collected from a standard normal distribution, such that

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-\frac{x^2}{2}} dx$$

- Then, $z\sqrt{t}$ is an observation of B_t.

Observations, cont.

- The GBM is given by $S_t = S_0 e^{(\mu - \frac{\sigma^2}{2})t + \sigma B_t}$, then

$$S_t = S_0 e^{(\mu - \frac{\sigma^2}{2})t + \sigma z \sqrt{t}}$$

  is an observation of S_t.

- Using the following MATLAB commands to collect standard normal observations,

```
z = normrnd(0,1);

z = randn;
```

Estimate of expectation, cont.

- Let z_i, $i = 1, \cdots, M$ be M independent observations collected from a standard normal distribution, such that

$$P(Z \leqslant z_i) = \Phi(z_i) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z_i} e^{-\frac{x^2}{2}} dx$$

  where $Z \sim N(0, 1)$ is a random variable.

- Then we have
  - $\mathbb{E}[B_t] \approx \frac{\sqrt{t}}{M} \sum_{i=1}^{M} z_i$
  - $\mathbb{E}[S_t] \approx \frac{S_0}{M} \sum_{i=1}^{M} e^{(\mu - \frac{\sigma^2}{2})t + \sigma z_i \sqrt{t}}$

Error analysis, $\mathbb{E}[B_t]$

- Recall that $\mathbb{E}[B_t] = 0$.
- We can obtain the following estimates of $\mathbb{E}[B_t]$,

t	M	Estimates	CPU time (s)
5	10^4	0.0518	0.091144
	10^5	-0.0101	0.829603
	10^6	0.0013	8.400436
	10^7	0.000994	82.361601
50	10^4	-0.0108	0.089226
	10^5	0.0095	0.863663
	10^6	0.0011	8.340475
	10^7	0.000463	83.029587
500	10^4	0.0893	0.087249
	10^5	-0.0054	0.811274
	10^6	0.0026	8.338143
	10^7	-0.000732	82.605615

Error analysis, $\mathbb{E}[S_t]$

- Recall that $\mathbb{E}[S_t] = S_0 e^{\mu t}$.
- Assume $S_0 = 1$, $\mu = 0.2$ and $\sigma = 0.5$, we can obtain the following estimates of $\mathbb{E}[S_t]$,

t	$\mathbb{E}[S_t]$	M	Estimates	CPU time (s)
5	2.7183	10^4	2.7318	0.092150
		10^5	2.7205	0.910633
		10^6	2.7194	8.555989
		10^7	2.7176	102.599107
6	3.3201	10^4	3.3948	0.095045
		10^5	3.3273	0.859025
		10^6	3.3224	8.284265
		10^7	3.3210	93.565865
7	4.0552	10^4	4.0879	0.086303
		10^5	4.0734	0.820039
		10^6	4.0618	8.478058
		10^7	4.0599	98.005538

Monte-Carlo simulation

- Let $f(t, B_t)$ be a stochastic process, then we have

$$\begin{aligned}\mathbb{E}[f(t, B_t)|\mathcal{F}_s] &= \mathbb{E}[f(t, B_s + (B_t - B_s))|\mathcal{F}_s] \\ &= \mathbb{E}[f(t, B_s + Z\sqrt{t-s})|\mathcal{F}_s]\end{aligned}$$

  where $\{\mathcal{F}_s, 0 \leq s < t\}$ is the filtration and $Z \sim N(0,1)$ is a random variable.

- By Monte-Carlo simulation,

$$\mathbb{E}[f(t, B_t)|\mathcal{F}_s] \approx \frac{1}{M}\sum_{i=1}^{M} f(t, B_s + z_i\sqrt{t-s})$$

Monte-Carlo simulation, cont.

- Let $S_t = S_0 e^{at+\sigma B_t}$ be a GBM, then we have

$$\begin{aligned}\mathbb{E}[f(S_T)|\mathcal{F}_t] &= \mathbb{E}\left[f\left(S_t e^{a(T-t)+\sigma(B_T-B_t)}\right)\middle|\mathcal{F}_t\right] \\ &= \mathbb{E}\left[f\left(S_t e^{a(T-t)+\sigma Z\sqrt{T-t}}\right)\middle|\mathcal{F}_t\right]\end{aligned}$$

  where $\{\mathcal{F}_t, 0 \leq t < T\}$ is the filtration and $Z \sim N(0,1)$ is a random variable.

- By Monte-Carlo simulation,

$$\mathbb{E}[f(S_T)|\mathcal{F}_t] \approx \frac{1}{M}\sum_{i=1}^{M} f\left(S_t e^{a(T-t)+\sigma z_i\sqrt{T-t}}\right)$$

Monte-Carlo simulation in option pricing

- Let $C = f(S_T)$ be the option claim,

$$g(t, S_t) = e^{-\rho(T-t)}\mathbb{E}[f(S_T)|\mathcal{F}_t]$$
$$\approx e^{-\rho(T-t)}\frac{1}{M}\sum_{i=1}^{M} f\left(S_t e^{a(T-t)+\sigma z_i\sqrt{T-t}}\right)$$

- The no-arbitrage condition is still $a = \rho - \frac{\sigma^2}{2}$.

European options

- European call option:

$$g(t, S_t) = e^{-\rho(T-t)}\mathbb{E}[(S_T - K)_+ | \mathcal{F}_t]$$
$$\approx e^{-\rho(T-t)} \frac{1}{M} \sum_{i=1}^{M} \left(S_t e^{a(T-t)+\sigma z_i \sqrt{T-t}} - K \right)_+$$

- European put option:

$$g(t, S_t) = e^{-\rho(T-t)}\mathbb{E}[(K - S_T)_+ | \mathcal{F}_t]$$
$$\approx e^{-\rho(T-t)} \frac{1}{M} \sum_{i=1}^{M} \left(K - S_t e^{a(T-t)+\sigma z_i \sqrt{T-t}} \right)_+$$

Example

- The current price of a non-dividend paying share is £14 and its volatility is thought to be 25% per annum. The continuously compounded risk-free interest rate is 4.8% per annum. A European option on this share has a strike price of £18 and term to maturity of half year. Assume that the Black-Scholes model applies.
- $S_0 = 14$, $K = 18$, $T = 0.5$, $\sigma = 0.25$, $\rho = 0.048$, we calculate

$$d_1 = \frac{\log\left(\frac{S_0}{K}\right) + \left(\rho + \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}} = -1.1975$$

$$d_2 = d_1 - \sigma\sqrt{T} = -1.3743$$

Example, cont.

- If the option is a European call option, then by Black-Scholes model the time 0 option price is:

$$g(0, S_0) = S_0 \Phi(d_1) - K e^{-\rho T} \Phi(d_2) = 0.1297$$

- Then by Monte Carlo,

$$g(0, S_0) \approx e^{-\rho T} \frac{1}{M} \sum_{i=1}^{M} \left(S_0 e^{\left(\rho - \frac{\sigma^2}{2}\right) T + \sigma z_i \sqrt{T}} - K \right)_+$$

Example, cont.

- If the option is a European put option, then by Black-Scholes model the time 0 option price is:

$$g(0, S_0) = Ke^{-\rho T}\Phi(-d_2) - S_0\Phi(-d_1) = 3.7029$$

- Then by Monte Carlo,

$$g(0, S_0) \approx e^{-\rho T}\frac{1}{M}\sum_{i=1}^{M}\left(K - S_0 e^{\left(\rho - \frac{\sigma^2}{2}\right)T + \sigma z_i\sqrt{T}}\right)_+$$

Error analysis

- Results are listed below,

Options	$g(0, S_0)$	M	Estimates	CPU time (s)
Call	0.1297	10^4	0.1312	0.002365
		10^5	0.1304	0.006900
		10^6	0.1302	0.064002
		10^7	0.1298	0.637134
		10^8	0.1297	6.442940
Put	3.7029	10^4	3.7449	0.002414
		10^5	3.7120	0.006709
		10^6	3.7070	0.064126
		10^7	3.7032	0.665505
		10^8	3.7027	6.545800

Examples

- State how to use Monte-Carlo method to calculate the time t option prices for the following exotic options.
  - $f(S_T) = \max\left(|S_T|, \sin(S_T^3)\right)$.
  - $f(S_T) = \cos[\log(S_T^3)]$.
  - $f(S_T) = \frac{1}{3+S_T^2}$.